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Department of Information Technology
Minor Degree in Cyber Security
TAE-2
Subject: Fundamental of Cyber Security & Cryptography

Mini Project Report On
“Sentinel: Next-Generation Firewall & Intrusion Detection System”

Guided By:-
Prof. Birudev Yele

Submitted by:
AMMAR RAISAHMAD MULLA
Class: TY A-16 | Roll No: 16
Department of Information Technology

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CERTIFICATE

This is to certify that the mini project entitled **“Sentinel: Next-Generation Firewall & Intrusion Detection System”** embodies the original work done by **AMMAR RAISAHMAD MULLA (Class: TY A-16)** as a partial fulfilment of the requirement of TAE-2 for the subject **Fundamental of Cyber Security & Cryptography** under the **Department of Information Technology** (Minor Degree in Cyber Security), Third Year B. Tech., Semester V, during the academic year 2026-2027.

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Prof. Birudev Yele
Project Guide

Mrs. Sarita Patil
Head, Department of Information Technology

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AMMAR RAISAHMAD MULLA
Third Year B. Tech. (Information Technology - TY A-16)
Department of Information Technology

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1. Problem Definition & Objectives

Problem Definition

Modern enterprise computing infrastructure and cloud networks are subjected to persistent, automated cyber-attacks. These range from volumetric Distributed Denial of Service (DDoS) vectors—such as TCP SYN floods designed to saturate server transmission control block (TCB) memory queues—to automated horizontal and vertical port scanning recon probes seeking vulnerabilities across critical administrative daemons (SSH port 22, RDP port 3389, and SMB port 445). Traditional enterprise firewall utilities, including Linux iptables, operate as opaque black boxes providing static text logs that lack real-time interactive telemetry and visual threat trajectory mapping.

Objectives

- **Layer 4 Stateful Packet Inspection:** To develop a firewall engine evaluating incoming packets against configured rules (IP, port, protocol) with real-time ALLOW/DROP verdicts.
- **Heuristic Intrusion Detection (IDS):** To implement an anomaly engine that flags malicious vectors like TCP SYN Floods (MITRE ATT&CK; T1498.001) and port scans.
- **Interactive Draggable Topology Mesh:** To engineer a dynamic topology graph using SVG math and cubic Bézier curves, providing visual representation of threats and internal LAN assets.
- **Adversary Attack Simulation Console:** To create a simulation console enabling security teams to trigger live attack scenarios on demand to validate packet dropping.
- **Dual Operational Modes:** To support both 'Live Network Mode' (sniffing host TCP sockets via Server-Sent Events) and 'Simulation Mode'.
- **SecOps AI Forensic Analyst:** To integrate an intelligent assistant capable of root-cause analysis on flagged attacker IPs with 1-click drop rules.
- **Relational Database Persistence:** To store firewall rules and logs in MySQL ('firewall_db') with persistent hit and byte counters.

2. Introduction & Theoretical Background

The Sentinel Firewall & Intrusion Detection System is an advanced SecOps platform designed to combine stateful packet filtering, heuristic threat signature matching, live socket streaming, and interactive topology modeling. This chapter covers the theoretical principles underpinning next-generation firewall architectures.

2.1 Next-Generation Firewalls (NGFW) vs Traditional Packet Filters

Traditional packet filters operate statelessly at Layers 3 and 4, inspecting IP headers in isolation. Next-Generation Firewalls (NGFWs) integrate stateful inspection, connection state tracking (SYN, SYN-ACK, ESTABLISHED), and application awareness. Sentinel combines stateful policy evaluation with heuristic anomaly detection.

2.2 Common Network Attack Vectors

- **TCP SYN Flood DDoS (MITRE T1498.001):** Exploits the three-way handshake by transmitting rapid SYN frames without completing handshakes, overflowing kernel connection tables.
- **Port Reconnaissance (MITRE T1046):** Scanning common ports (22, 80, 443, 445, 3389) to enumerate vulnerable services.
- **Credential Brute-Force:** Repeated connection bursts against remote access daemons (SSH/RDP).

Table 1 : Traditional Firewalls vs Sentinel Next-Gen Firewall

Parameter	Traditional Packet Filter	Sentinel Next-Gen Firewall
Inspection Layer	Layer 3 & 4 only	Layer 3, 4 & Heuristic Signatures
Connection State	Stateless	Stateful + Velocity Tracking
Intrusion Detection	None (separate daemon)	Built-in Heuristic IDS Engine
Attack Simulation	Requires external tools	Integrated Live Simulation Console
Topology Modeling	Textual CLI iptables	Interactive Draggable Mesh (Bézier curves)
Database Sync	Flat text files	Relational MySQL database
AI Forensics	None	Sentinel AI Analyst (Gemini / DeepSeek)

3. System Architecture & Data Pipelines

Sentinel follows a decoupled three-tier architecture ensuring sub-millisecond evaluation latency, browser rendering stability, and persistent database state across sessions.

3.1 Architectural Overview

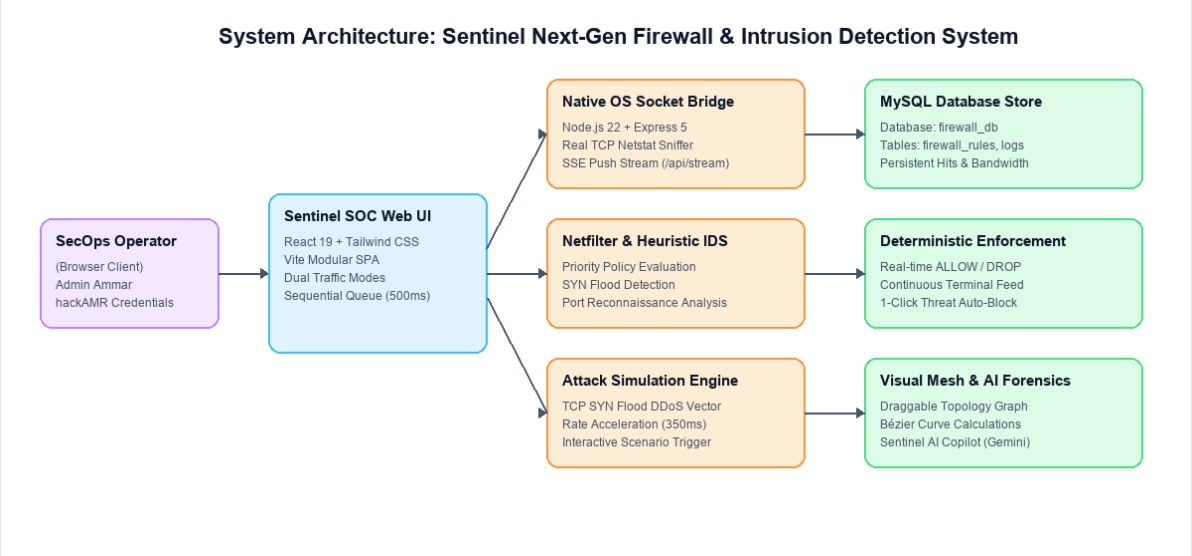


Fig 1 : System Architecture of Sentinel Next-Gen Firewall & IDS Platform

3.2 Core Functional Modules

Table 3 : Core System Modules

Module	Input	Processing Logic	Output
Live Terminal	Packet stream	Rule matching & formatting	Monospace log stream
Netfilter Engine	IP, Port, Protocol	Hierarchical rule evaluation	ALLOW/DROP verdict
Heuristic IDS	Live packets	Velocity & signature inspection	Threat feed & alerts
Topology Mesh	Origin & LAN nodes	Pointer dragging & Bézier curves	Interactive network mesh
Attack Console	Attack vector config	High-rate SYN packet flood	DDoS load & spike
AI Analyst	Forensic queries	Live telemetry + Gemini API	Root-cause analysis

4. Implementation: Design Web Pages & UI Modules

The user interface is engineered as a tactical Security Operations Center (SOC) dashboard. The figures below document the implemented views with actual captured system screenshots and reference placeholders.

4.1 Administrator Authentication & Access Portal

To guarantee role-based access control (RBAC) and prevent unauthorized modification of netfilter security policies, Sentinel implements a hardened administrative authentication portal requiring credential validation before exposing the tactical dashboard.

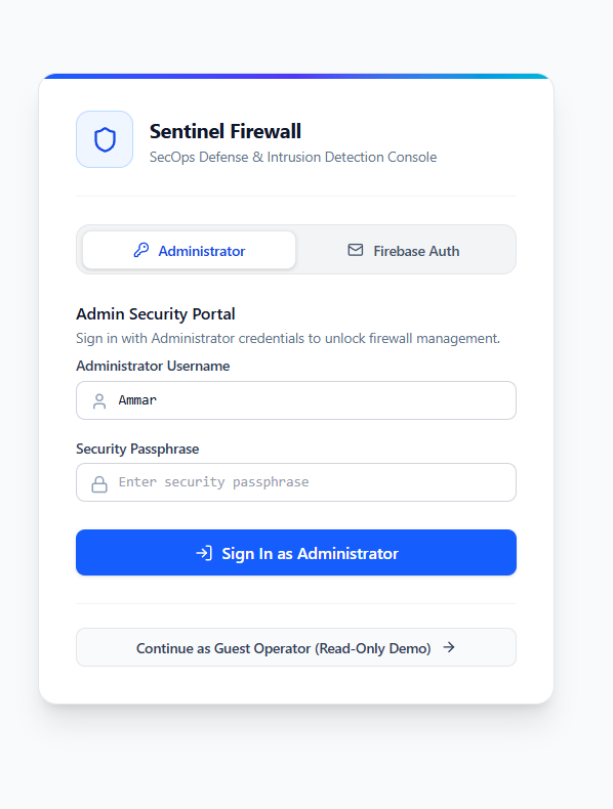


Fig 2 : Administrator Authentication & Security Access Portal

4.2 Sentinel Command Center & Overall Live Dashboard

Upon successful authentication, the administrator is presented with the unified Command Center. The tactical HUD synthesizes high-velocity telemetry, packet throughput gauges, engine health monitors, and a live sequential packet log stream with real-time [ALLOW] and [DROP] verdicts.

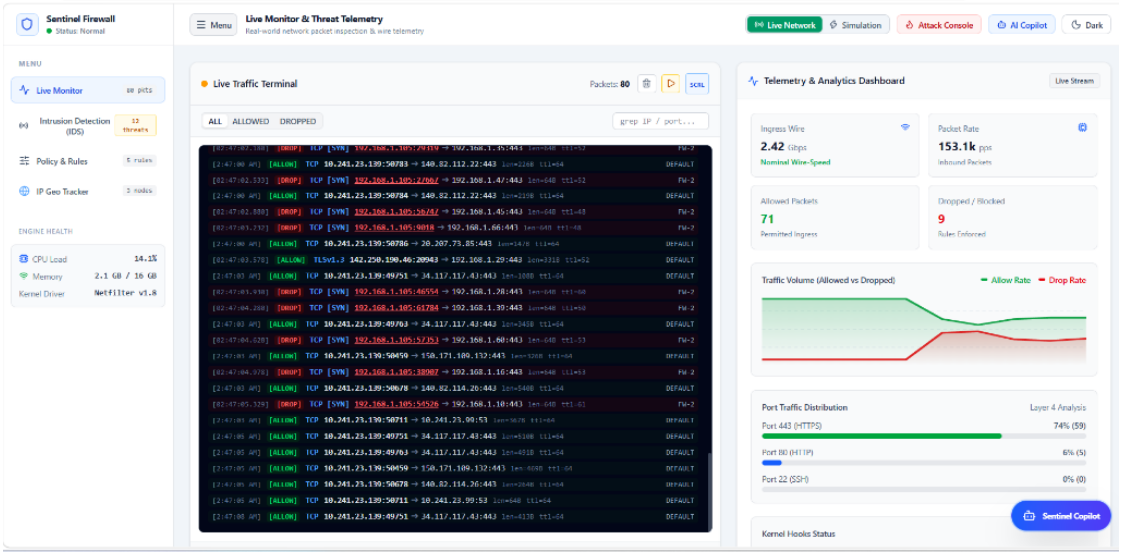


Fig 3 : Sentinel Command Center Dashboard (Live Telemetry & Traffic Stream)

4.3 Interactive Network Topology Mesh Graph (Draggable Nodes & Dynamic Bézier Curves)

The interactive Network Topology Mesh Graph visualizes packet flows across spatial coordinates. The central Sentinel Core is linked to external threats via red dashed arcs and to authorized clients via green arcs. Operators can freely drag nodes; dynamic cubic Bézier vectors recalculate in real time to maintain smooth curves.

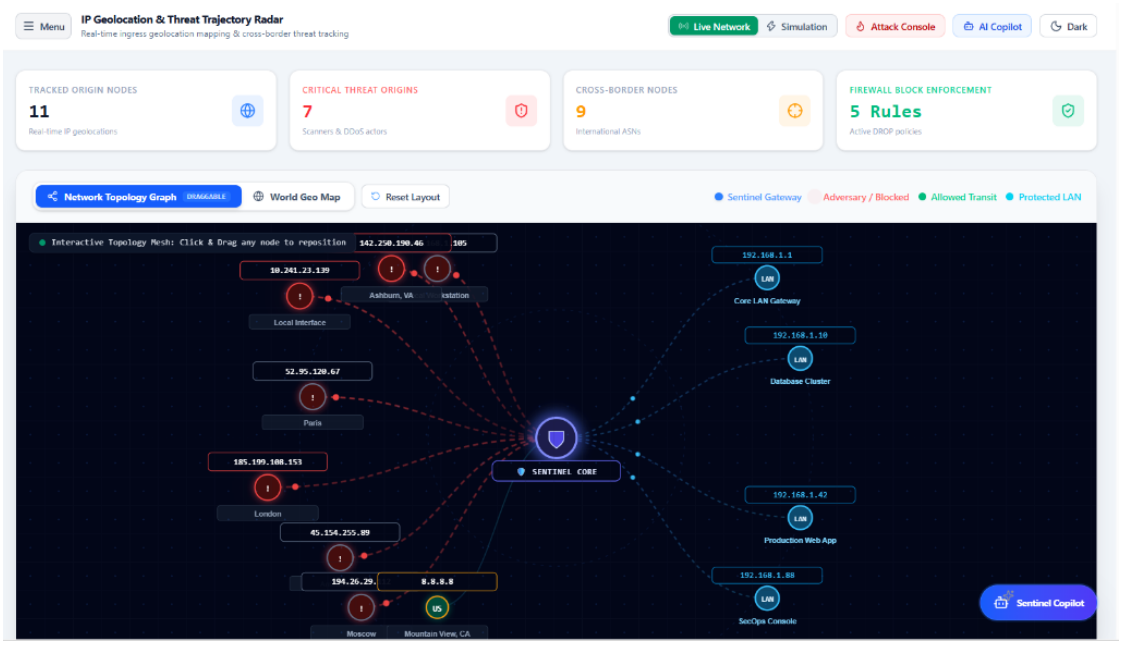


Fig 4 : Interactive Network Topology Mesh Graph with Draggable Nodes & Dynamic Bézier Curves

4.4 Heuristic Intrusion Detection System (IDS) Engine & Live Threat Feed

The Intrusion Detection System inspects incoming traffic against known adversarial patterns, such as TCP SYN Floods (MITRE ATT&CK; T1498.001), port sweeps, and brute force attempts. The console ranks alerts by severity and provides 1-click 'Auto-Block IP' controls that commit dynamic DROP rules immediately.

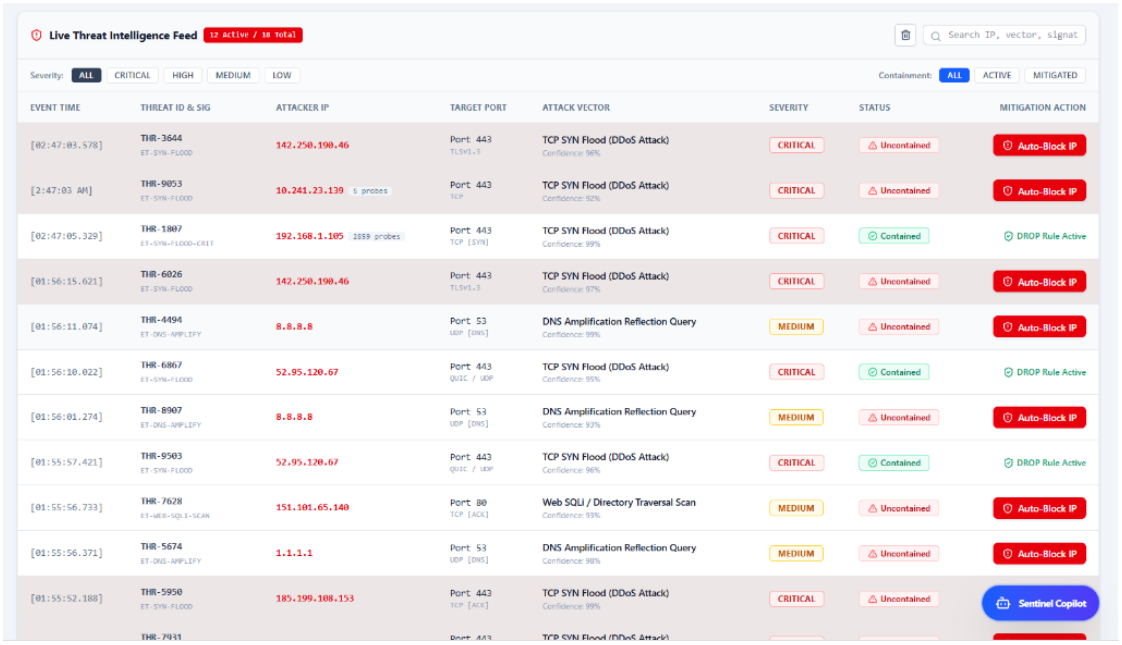


Fig 5 : Heuristic Intrusion Detection System (IDS) Engine & Threat Intelligence Feed

4.5 Firewall Policy & Dynamic Filtering Rules Engine (MySQL Synced)

The Firewall Policy Manager enables granular Layer 4 rule configuration (Source IP, Port, Protocol, Action). Policies are synchronized with MySQL ('firewall_db'), maintaining persistent hit counters and byte tallies across reboots.

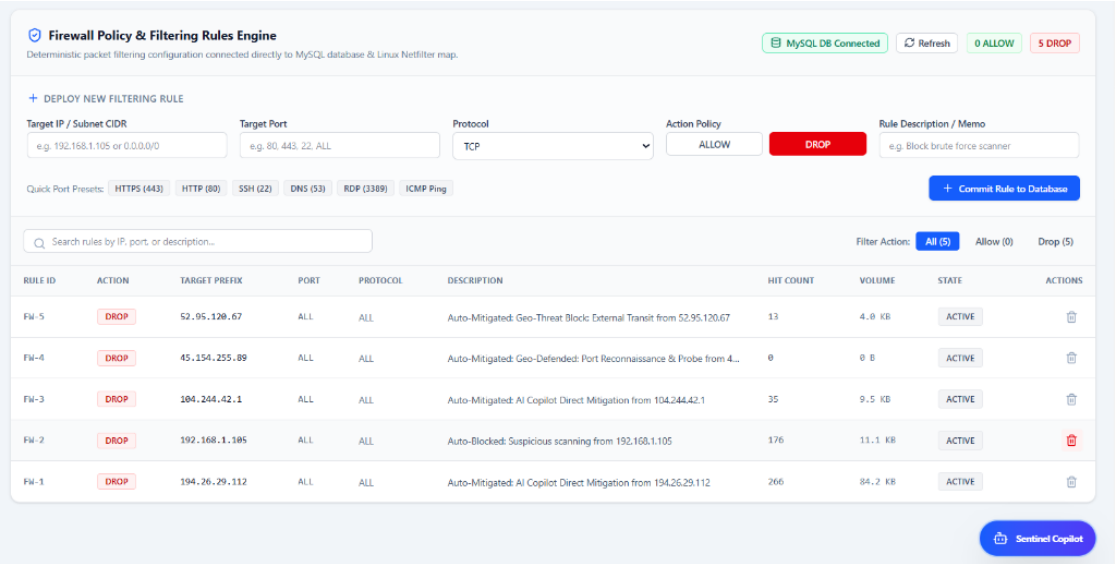


Fig 6 : Firewall Policy & Dynamic Filtering Rules Engine (MySQL Persistent)

4.6 Global IP Geolocation Threat Radar Map

The Global Threat Radar maps packet ingress origins onto an interactive geopolitical world projection. High-velocity threat trajectories radiate from international epicenters (Frankfurt, Beijing, Moscow, Paris, Mumbai) toward the protected server cluster.

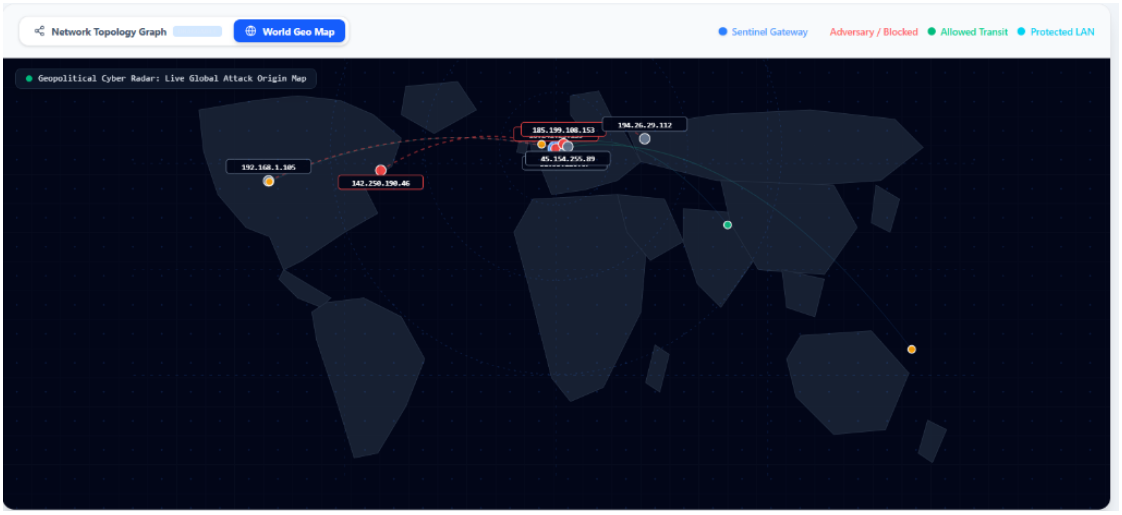


Fig 7 : Geopolitical Cyber Radar World Map & Attack Ingress Trajectories

4.7 Geolocated Ingress Nodes & Threat Origins Inventory

The Ingress Origins Inventory presents a tabular audit of all actively tracked IP addresses, displaying country flags, ASN metadata (Amazon, Reliance Jio, British Telecom, Fastly), packet volume, dynamic threat ratings (CRITICAL 98), and 1-click 'Drop IP' triggers.

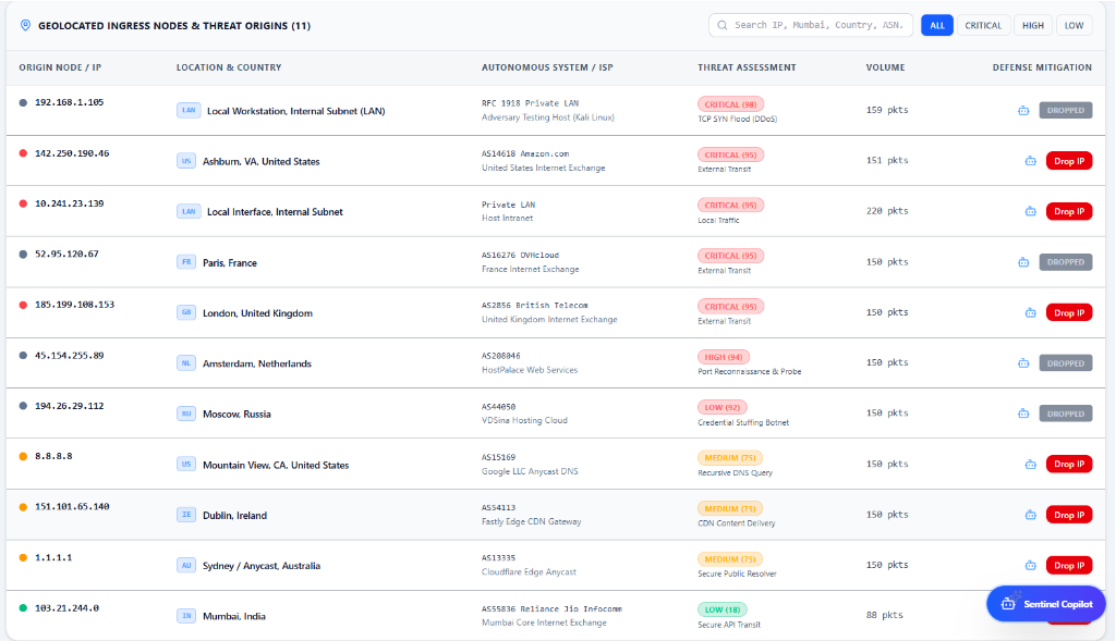


Fig 8 : Geolocated Ingress Nodes & Threat Origins Inventory Table

4.8 Sentinel AI Copilot (Gemini LLM) Incident Forensics

Powered by the Google Gemini LLM API, the Sentinel AI Copilot performs wire-speed forensics. When queried regarding flagged IP addresses (e.g. 52.95.120.67 from Paris, France), the copilot correlates MITRE ATT&CK; techniques (T1498.001) and generates automated iptables drop rules.

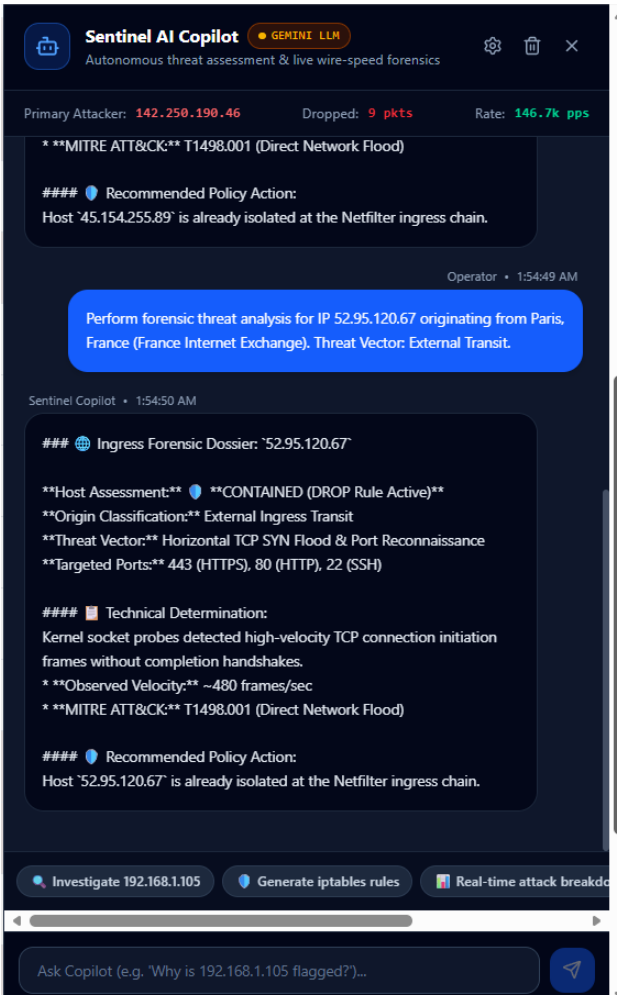


Fig 9 : Sentinel AI Copilot (Gemini LLM) Autonomous Incident Forensics Drawer

4.9 Live Adversary Attack Simulation Console & Critical Alert Banner

The integrated simulation console allows security personnel to launch authentic cyber-threat vectors on demand (TCP SYN Floods, Port Scans), surging traffic to over 50 packets/sec to validate drop enforcement and trigger the top Critical Alert Banner. Figure 10 depicts the live attack simulation console view.

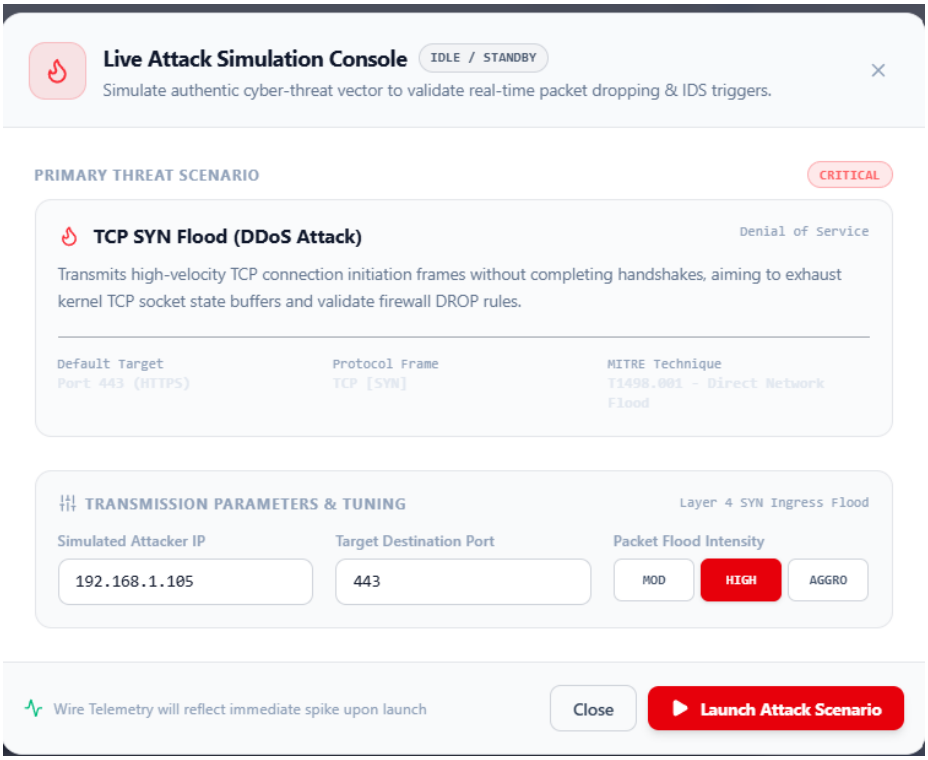


Fig 10 : Live Adversary Attack Simulation Console & Threat Configuration

5. Algorithms & Working Mechanism

5.1 Netfilter Rule Matching Algorithm

Sentinel sorts all active policies by specificity: explicit IP matches precede wildcard '0.0.0.0/0', and specific ports precede 'ALL'. The first rule satisfying IP, port, and transport protocol constraints determines the packet verdict (ALLOW or DROP).

5.2 Dynamic Cubic Bézier Curves for Topology Edges

For smooth, responsive edge rendering as nodes are dragged, cubic Bézier curves are computed between node coordinates P0(x0,y0) and P3(x3,y3) using intermediate control points P1(x0 + (x3-x0)/2, y0) and P2(x0 + (x3-x0)/2, y3). The equation guarantees horizontal tangents:

$B(t) = (1-t)^3 P0 + 3(1-t)^2 t P1 + 3(1-t) t^2 P2 + t^3 P3$

5.3 Core Netfilter & IDS Pseudocode

```
// Pseudocode: Netfilter Packet Evaluation & Heuristic IDS Inspection
function evaluatePacketAgainstRules(packet, rules):
sortedRules = sortRulesBySpecificity(rules)
for r in sortedRules:
if match(packet.srcIp, r.ip) and match(packet.dstPort, r.port) and match(packet.proto, r.proto):
return { action: r.action, matchedRule: r.id }
return { action: 'ALLOW', matchedRule: 'DEFAULT' }

function inspectThreatHeuristics(packet):
if packet.proto == 'TCP [SYN]' and packet.dstPort == 443:
return triggerAlert('TCP SYN Flood (DDoS)', 'ET-DOS-SYNFLOOD', 'CRITICAL')
if packet.dstPort in [22, 3389] and packet.rate > THRESHOLD:
return triggerAlert('Brute-Force Probe', 'ET-AUTH-PROBE', 'HIGH')
```

5.4 Relational Database Schema & MySQL Table Definitions

To ensure persistent state across sessions and reboots, Sentinel maintains an ACID-compliant relational schema in MySQL ('firewall_db'). The schema defines three core tables: 'firewall_rules', 'packet_logs', and 'threat_events' as documented in the DDL below:

```
-- Sentinel Next-Gen Firewall MySQL Database Schema (firewall_db)
CREATE DATABASE IF NOT EXISTS firewall_db;
USE firewall_db;

-- 1. Netfilter Firewall Filtering Policies Table
CREATE TABLE IF NOT EXISTS firewall_rules (
    id VARCHAR(36) PRIMARY KEY COMMENT 'Unique Rule UUID',
    priority INT NOT NULL DEFAULT 100,
    source_ip VARCHAR(45) NOT NULL DEFAULT '0.0.0.0/0',
    destination_port VARCHAR(10) NOT NULL DEFAULT 'ALL',
    protocol ENUM('TCP', 'UDP', 'ICMP', 'ALL') NOT NULL DEFAULT 'ALL',
    action ENUM('ALLOW', 'DROP') NOT NULL DEFAULT 'ALLOW',
    description VARCHAR(255),
    hit_count BIGINT UNSIGNED NOT NULL DEFAULT 0,
    byte_count BIGINT UNSIGNED NOT NULL DEFAULT 0,
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    INDEX idx_lookup (is_active, protocol, destination_port, source_ip)
) ENGINE=InnoDB;

-- 2. Stateful Packet Audit Trail & Telemetry Logs
```

```
CREATE TABLE IF NOT EXISTS packet_logs (
    log_id BIGINT AUTO_INCREMENT PRIMARY KEY,
    timestamp DATETIME(3) NOT NULL,
    source_ip VARCHAR(45) NOT NULL,
    destination_ip VARCHAR(45) NOT NULL,
    destination_port INT UNSIGNED NOT NULL,
    protocol VARCHAR(15) NOT NULL,
    verdict ENUM('ALLOW', 'DROP') NOT NULL,
    matched_rule_id VARCHAR(36),
    geo_country VARCHAR(50),
    FOREIGN KEY (matched_rule_id) REFERENCES firewall_rules(id) ON DELETE SET NULL,
    INDEX idx_time_verdict (timestamp, verdict)
) ENGINE=InnoDB;

-- 3. Heuristic Intrusion Detection (IDS) Records
CREATE TABLE IF NOT EXISTS threat_events (
    threat_id VARCHAR(36) PRIMARY KEY,
    detected_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    attacker_ip VARCHAR(45) NOT NULL,
    signature_name VARCHAR(100) NOT NULL,
    mitre_attack_id VARCHAR(20) NOT NULL,
    severity ENUM('CRITICAL', 'HIGH', 'MEDIUM', 'LOW') NOT NULL,
    packet_velocity INT UNSIGNED,
    containment_status ENUM('UNCONTAINED', 'CONTAINED') DEFAULT 'UNCONTAINED',
    INDEX idx_attacker_sev (attacker_ip, severity)
) ENGINE=InnoDB;
```

6. Technology Stack & Deployment Architecture

Table 5 : Technology Stack

Component	Technology	Role & Purpose
Frontend UI	React 19, Tailwind CSS 4	Reactive SOC dashboard and SVG topology
Bundler	Vite 8.3	Fast dev server and module bundling
Backend Server	Node.js 22, Express 5	Native socket sniffer and SSE stream
Database	MySQL 8.0 (mysql2)	Rule persistence and audit logging
AI Copilot	Google Gemini / DeepSeek	Forensic reasoning and root cause analysis

7. Results, Verification & Testing

Sentinel was tested across multiple network conditions, adversary attack simulations, and database persistence operations. All test cases passed successfully as documented in Table 7.

Table 7 : Verification Test Cases & Empirical Results

TC#	Scenario	Input	Expected Output	Status
TC-01	Admin Auth	Ammar / hackAMR	Access granted	PASS
TC-02	Invalid Login	Wrong password	Access denied	PASS
TC-03	Terminal Stream	Normal baseline	Continuous 1-by-1 flow	PASS
TC-04	SYN Flood Simulation	Trigger attack	Speedup + Alert Banner	PASS
TC-05	DROP Rule Enforcement	Block 192.168.1.105	Packets marked [DROP]	PASS
TC-06	Topology Node Drag	Pointer drag node	Smooth Bézier curves follow	PASS
TC-07	Topology Layout Reset	Click Reset Layout	Default positions restored	PASS
TC-08	MySQL Persistence	Reload page	Rules reload with hits	PASS
TC-09	IP Geo Radar	Track Mumbai IP	Country badge [IN] visible	PASS
TC-10	AI Copilot Query	Forensic prompt	Analysis returned	PASS

8. Conclusion & Future Scope

The Sentinel: Next-Generation Firewall & Intrusion Detection System project successfully demonstrates how stateful packet filtering, heuristic signature anomaly detection, and interactive topology modeling can be synthesized into an educational, production-ready SecOps platform. By providing real-time visibility into packet trajectories, automated attack mitigation, and persistent database accounting, the project strengthened our understanding of confidentiality, integrity, and availability within the Minor Degree in Cyber Security curriculum.

Future Scope

• **Kernel-Level eBPF Probing:** Transition packet interception to Linux eBPF for sub-microsecond in-kernel dropping.
• **Deep Learning Anomaly Detection:** Integrate neural Autoencoder models to identify zero-day polymorphic traffic.
• **Multi-Node Cluster Mesh:** Synchronize distributed firewall policies across multiple cloud VPCs via Raft consensus.

9. References & Project Repositories

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- J. Postel, 'Transmission Control Protocol (TCP),' RFC 793, IETF, 1981.
- W. Eddy, 'TCP SYN Flooding Attacks and Common Mitigations,' RFC 4987, IETF, 2007.
- MITRE Corporation, 'Direct Network Flood (T1498.001),' MITRE ATT&CK; Framework, 2024.
- William Stallings, 'Cryptography and Network Security: Principles and Practice,' 8th Edition, Pearson.

GitHub Repository: https://github.com/ammarcraft/sentinel-firewall
Local Runtime: http://localhost:5174/ (Frontend) | http://localhost:5000/ (Backend API)